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Entropy based Binary Particle Swarm Optimization and classification for ear detection

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ABSTRACT

Ear detection in facial images under uncontrolled environments with varying occlusion, pose, background *and* lighting conditions is challenging. In this paper, we propose a novel technique, namely *Entropic Binary Particle Swarm Optimization* (EBPSO) which generates an entropy map, the highest value of which is used to localize the ear in a face image. Also, Dual Tree Complex Wavelet Transform (DTCWT) based background pruning is used to eliminate most of the background in the face image. This is achieved as a result of DTCWT highlighting the strong curves in the foreground. The resulting preprocessed image contains the salient facial features and prepares the ground for ear detection. The *Entropy based classifier* successfully demarcates the ear regions from other facial features, based on observed patterns of entropy. Experimental results show the promising performance of EBPSO for ear detection on four benchmark face databases: CMU PIE, Pointing Head Pose, Color FERET and UMIST.

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1. Introduction

Ears have long been considered as a potential source of biometric classification. Since the studies conducted by Iannarelli (1989), indicating rich structures and consistency of shape over a long period, it has drawn a vast amount of interest. The early development of ear shape and invariance to facial expressions are among the important reasons for considering them advantageous in biometrics.

A basic ear biometric system has three stages, namely detection, feature extraction and recognition. Modern recognition algorithms are developed based on the assumption that ears have been correctly segmented. This emphasizes the importance of having a robust ear detection technique. A compilation of available techniques and challenges is available in the survey conducted by Abaza et al. (2011). The first attempt by Burge and Burger (2000) in detecting the ear required a human input and was not a fully automated approach. Subsequent attempts focused on semi-autonomous implementations, such as those by Alvarez et al. (2005).

Islam et al. (2008) used a cascaded Adaboost technique based on Haar features for ear detection. This technique is popularly known as the Viola–Jones method. Wu et al. (2008) modified the original approach to reduce the complexity of the training phase by two orders of magnitude, which Abaza et al. (2010) applied to ear detection. Although the training phase of their approach was about 80 times faster than the original method, it took up to 8 h.

Yan and Bowyer (2006, 2007) developed an automatic ear biometric system, requiring the input of both 2D and 3D range data; this had immense computational overheads. Their dataset did not focus on real-time conditions.

Swarm intelligence (Eberhart and Shi, 2004) provides an alternative for solving several complex problems. They have been applied in various domains like Biomedicine (Karabogaa and Latifoglub, 2013), System Identification (Luitel and Venayagamoorthy, 2010) and so on. Particle Swarm Optimization (PSO) is an interesting variant of swarm intelligence. Its meta-heuristic nature coupled with its evolutionary approach gives it immense potential to adapt to a number of applications. Understanding their natural behavior and interspersing them with suitably processed datasets help reduce complexity.

Each of the previous implementations of ear detection systems had unaddressed issue(s), which are briefly highlighted in Section 2. These issues incited us to develop an autonomous and generic 2D ear detection algorithm with minimum training time, which can be deployed in realistic environments. For this purpose we have specifically chosen databases featuring the face prominently as opposed to ear-centric databases that were used in the previous ear detection literature.

The rest of the paper is organized as follows. Section 2 remarks on the prevalent issues in ear biometrics and briefly highlights the

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primary contributions with respect to the constraints prevalent in the current methods. Section 3 highlights the existing techniques that have been used in this paper. Section 4 describes the proposed algorithm in detail. In Section 5, we apply the proposed ear detection scheme on a variety of databases and the results are tabulated along with their corresponding graphical interpretations. Also, it provides a concise analysis of the algorithm duly illustrated by successful and failed detections from each of the databases. We conclude our work in Section 6.

2. Problem statement and key contributions

An overview of literature available in the domain of ear detection was highlighted in Section 1. In particular the early implementations required human intervention in order to operate successfully. Further, the systems required massive training periods to make them more robust. Newer systems were designed to accurately handle various effects on the ear image, such as occlusion, lighting, and this led to more overheads.

In addition, many other issues in tackling face databases influenced the development of our algorithm. The use of different types of face databases required multiple skin models to be employed to identify facial features. The presence of significant background in certain databases also posed a challenge in identifying the facial features. Apart from this, issues such as pose and illumination variations also had a profound impact. The proposed system was designed to address these issues and our key contributions are listed below:

1. *Novel EBPSO algorithm*: An entropy map is created with predefined window size, providing a rich feature space to analyze and locate the ear. The entropy map varies in size depending on the database and BPSO is used to optimize the entropy map. A new variant of BPSO algorithm is proposed, whose computations are based purely on the entropy map. The estimation of window size, used in the generation of entropy map, is an automated process which makes use of only a few images (about 10%) from a specific database.
2. *Entropy based classification*: Ear images possess entropy values which lie within a specific range. A threshold based on this entropy value is used to classify successful and failed detections. We found that non-ear segments have lesser 2-bit entropy values compared to ear-segments.
3. *DTCWT based background pruning*: The use of DTCWT yields wavelets at 6 angular orientations, each having a magnitude and a phase component. In this paper, the magnitudes of each of these orientations have been combined to provide a single image which highlights the strong curves in the foreground. A threshold is set to identify the occurrence of these curves and, using this, rest of the background is pruned away.

3. Preliminaries and related works

This section highlights the methods which are used commonly in detection. Particular emphasis is on techniques that process skin, handle edges and other features in the image.

3.1. Skin segmentation

With respect to its usage in ear biometrics, Prakash et al. (2009) suggested a hybrid technique to initially extract the skin region from image and then use this to obtain the ear using template matching. Chen and Bhanu (2005) proposed a technique to fuse skin color from color images and edges from range images, to perform ear detection. Similarly Yuan and Mu (2007) utilized a combination of skin-color and contour information to localize the ear. Yan and Bowyer (2007) introduced a new method which utilizes only color information to detect ear.

The purpose of using skin models is to localize the search within a probable skin region instead of the complete image. In this paper, skin models such as Hue, Saturation and Value (HSV) and Gaussian skin models have been used.

3.2. Dual Tree Complex Wavelet Transform

Discrete Wavelet Transform suffers from shortcomings like lack of directionality and shift variance around singularities. To overcome these drawbacks, Dual Tree *Complex* Wavelet Transform (DTCWT) was developed (Bodade and Talbar, 2011; Kingsbury, 1998; Selesnick et al., 2005). Due to the directionality of DTCWT the singularities are oriented in 6 angles which highlight all possible curvatures and hence can be utilized to detect and remove backgrounds in images. These properties inspired us to use DTCWT for background pruning which will be explained in Section 4.1.2.

3.3. Edge detection

Edge detection has been a dominant technique, employed consistently in ear detection; works of Yan and Bowyer (2006, 2007) employed an edge dependant technique to detect the ear pit based on nose position. Ansari and Gupta (2007) applied the canny edge detector on the complete image to segregate convex and concave edges while Chen and Bhanu (2005, 2007) used edge detection on range images to exploit the increased density around helix and anti-helix regions. Joshi and Chauhan (2011) experimented with separate edge and template matching based approaches to locate the ear region while Arbab-Zavar and Nixon (2007) applied the hough transform to enhance the elliptical features within the edge detected image. The essence of canny edge detection (Canny, 1986), in our work, is to utilize dense edges to detect the presence of ear features.

3.4. Entropy

The mathematical framework of entropy is provided by Information Theory, derived from the works of Shannon and Waver (1948). According to this, all information content can be modeled based on their probability of occurrence or non-occurrence. For example, a random event E with a probability of occurrence $P(E)$ is said to contain information, given by $-\log[P(E)]$.

If any source emits a discrete set of random variables given by $\{r_1, r_2, r_3, \ldots, r_k\}$ with probabilities $\{p_1, p_2, p_3, \ldots, p_k\}$, then the average information possessed by the random variables is defined as the *entropy* and is given by

$$H = -\sum_{i=1}^{k} p_i \log[p_i] \tag{1}$$

There have been very few references in previous literatures regarding the use of entropy for detection purposes. Wang et al. (2008) used the values of entropy to approximate the locations of the human eyes, which were then used to estimate the location of the person's mouth. Using these the facial features were detected. Khan et al. (2011) used entropy values of local regions to predict the facial expressions. In our work, we have further explored the characteristics of entropy and proposed the concept of an *entropy map*. This aided in developing a novel variant of the BPSO algorithm which was utilized to detect human ears in a given profile image.

3.5. Binary Particle Swarm Optimization

Particle Swarm Optimization is an optimization scheme developed by Kennedy and Eberhart (1995). PSO emphasizes on search space optimization which is used to find an optimum solution to a problem irrespective of search space. It comprises a set of randomly initialized particles, each posing a potential solution to the problem. These particles possess a stochastic memory, comprising their personal best and the best solutions amongst all the available particles. Their solutions are recursively evaluated using a fitness measure to obtain the best particle.

Particle Swarm Optimization scheme provides a platform for a variety of extensions. One popular extension is the Binary version of PSO called as *Binary Particle Swarm Optimization* (BPSO) (Kennedy and Eberhart, 1997). BPSO develops on the ideas of PSO while applying minor changes to the algorithm. In our work, we have proposed a variant of BPSO called the EBPSO, whose optimizations are based on the values of entropy present in an entropy map.

4. Proposed entropy based ear detection system

The proposed ear detection scheme was developed based on the fact that the ear region contains a rich depth of singularities (edges) when compared to the face region surrounding it. This aids in identifying one key region, of a unique size, which contains the ear. The process flow of the proposed ear detection system is shown in Fig. 1. The system is divided into the following stages:

1. *Preprocessing*: This stage of the algorithm seeks to neutralize the effect of background and focus on the subject's face. This helps to concentrate focus of the search.
2. *EBPSO algorithm and its application to ear localization*: The second stage involves the creation of an entropy map based on a standard window size, which is derived for each database separately. This is used to provide a possible search space to locate the ear.
3. *Entropy based classification*: A common trend in entropy values observed using empirical data, rendered its use as a simple, yet effective, classification method.

We describe each stage of the proposed system in detail in the following sections.

4.1. Preprocessing: face localization and background pruning

The various preprocessing techniques involved in the proposed ear detection algorithm are described below.

4.1.1. Skin segmentation and face localization

Skin segmentation technique similar to that proposed by Ghazali et al. (2011) is used here. Two separate binary images are obtained, one using HSV and the other using Gaussian Skin model. These images are then logically combined (with an AND operation) to obtain skin regions common in both skin models. The usage of two skin models allows the system to overcome falsely detected skin regions.

In a profile image of any subject, logic dictates that the facial region will contain maximum skin concentration. So, using the skin model, a search is performed to estimate the location of the face and thus eliminate the other regions; refer skin segmentation block in Fig. 1. Although simple logical operations can be used to detect the facial regions in images with lot of background information, in certain other databases, the images contain very few, and sometimes no background. In such cases a more accurate estimate can be obtained by performing a few more morphological operations on the image.

Region containing maximum skin concentration is segmented, and this region contains primary facial features of the subject which almost always includes the ear. This region is processed further to remove the remaining background.

4.1.2. Proposed DTCWT based background pruning

2D wavelets are highly oriented in nature and this orientation property is used in identifying singularities in an image. In case of 2D images *singularities translate to edges or curves*.

The DTCWT improvises on the DWT in that it provides both real and complex oriented wavelets. DTCWT produces six wavelets at $\pm 75^\circ$, $\pm 45^\circ$ and $\pm 15^\circ$. Each of these succeeds in isolating singularities occurring at their respective orientations. In this paper, we propose the use of both real and complex oriented wavelets to create images representing the magnitude and phase for each of the 6 angular orientations. The mathematical basis (Selesnick et al., 2005) for obtaining these is discussed below.

Consider a two dimensional image plane $I(x,y)$, which is the region with maximum skin concentration obtained at the output of face localization; it is represented by block ① in Fig. 2. In $I(x,y)$, x-axis represents rows and y-axis represents the columns in the

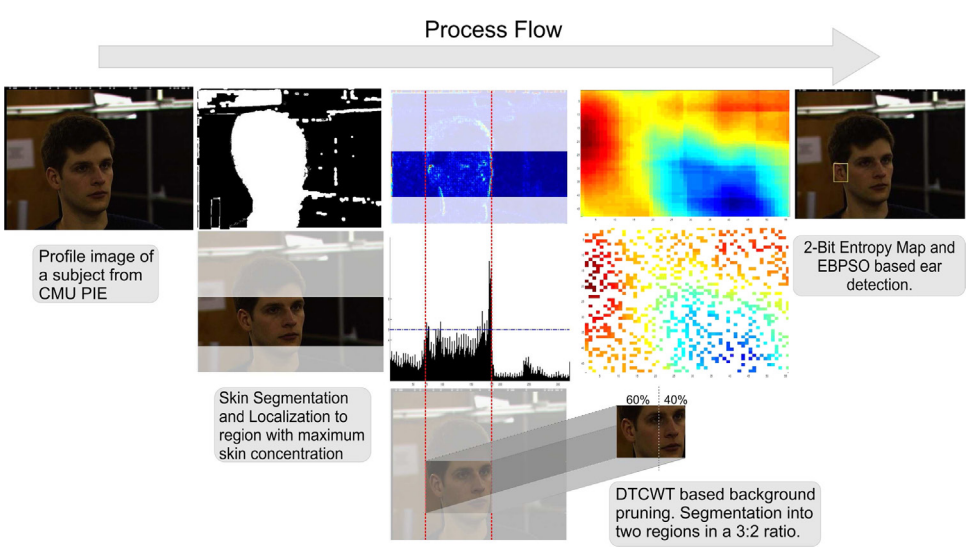

Fig. 1. Proposed ear detection system.

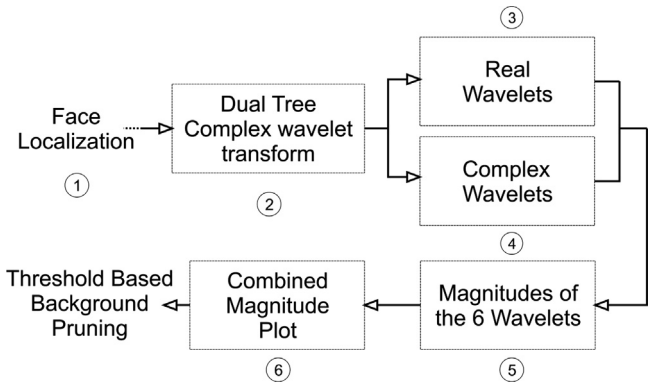

Fig. 2. Process flow of DTCWT based background pruning.

image. To begin with we assume that $\psi()$ represents the high pass operation and $\phi()$ represents the low pass operation on the image $I(x,y)$. Then, DTCWT wavelets oriented at -45° can be expressed as

$$\psi(x,y) = \psi(x).\psi(y) \tag{2a}$$

Here, $\psi(x)$ is the row implementation of the complex wavelet transform and $\psi(y)$ is the column implementation. These can be expanded as

$$\psi(x) = \psi_h(x) + j\psi_g(x) \tag{2b}$$

$$\psi(y) = \psi_h(y) + j\psi_g(y) \tag{2c}$$

From Eqs. (2a), (2b) and (2c) we have

$$\psi(x,y) = (\psi_h(x) + j\psi_g(x)) \cdot (\psi_h(y) + j\psi_g(y)) \tag{2d}$$

Eqs. (2a)–(2d) constitute a part of block ② in Fig. 2. The real and imaginary parts of Eq. (2d) are, respectively, given by

$$\mathrm{Re}\{\psi(x,y)\} = \psi_h(x) \cdot \psi_h(y) - \psi_g(x) \cdot \psi_g(y) \tag{2e}$$

$$\mathrm{Im}\{\psi(x,y)\} = \psi_g(x) \cdot \psi_h(y) + \psi_h(x) \cdot \psi_g(y) \tag{2f}$$

These equations are represented by blocks ③ and ④ in Fig. 2. Also, it is worth noting that the subscripts '*g*' and '*h*' represent the filter coefficients, which form an approximate Hilbert transform pair. From Eqs. (2e) and (2f) the magnitude (denoted by ρ_{-45°) for this orientation is given by

$$\rho_{-45^\circ}(x,y) = [(\psi_h(x) \cdot \psi_h(y) - \psi_g(x) \cdot \psi_g(y))^2 + (\psi_g(x) \cdot \psi_h(y) + \psi_h(x) \cdot \psi_g(y))^2)]^{1/2} \tag{3a}$$

This constitutes block ⑤ in Fig. 2. Solving the above equations we get

$$\rho_{-45^\circ}(x,y) = |\psi(x)| \cdot |\psi(y)| \tag{3b}$$

It is possible to obtain the other orientations in a similar manner. For example, the product $\psi(x) \cdot \overline{\psi(y)}$ (Note: $\overline{\psi(y)}$ is the complex conjugate of $\psi(y)$) which gives orientations at $+45^\circ$ and others can be obtained using $\phi(x)\psi(y)$, $\psi(x)\phi(y)$, $\phi(x)\overline{\psi(y)}$ and $\psi(x)\overline{\phi(y)}$. Fig. 3 shows the magnitudes of these wavelets oriented at 6 different angles.

The proposed technique generates 6 magnitude plots of singularities present at various orientations. A new *combined magnitude plot* is then created by adding the magnitudes of all 6 orientations (this constitutes block ⑥ of Fig. 2). This combined magnitude plot, shown in Fig. 3, contains the information of the singularities present in all orientations. Since this image is a combination of magnitudes at all orientations, it is *effective in suppressing singularities* that are not prominent in all orientations. The application of 1-level DTCWT on an image produces a resultant image half the size of the original.

The concept of *threshold based facial feature identification* is utilized to crop the resultant image so as to retrieve the facial features. All column elements of the matrix representing the combined magnitude plot are summed creating a 1-D row vector, hereafter called *Effective Row Vector* (ERV), in order to help identify the occurrence of the facial features (Fig. 4b). The values held have a direct relationship with the singularities present in a column.

From the combined magnitude plot we see that important facial features such as the nose-tip, chin or eyebrows possessed markedly high magnitudes owing to their multi-directional singularities and there was a sharp change in magnitude where they did not exist (refer Fig. 4b). Also they were prominent over the background.

A threshold was set on the values of the ERV. This value was determined experimentally using few images from each database. The threshold values for any image in a database always lie within a small range. A static unique threshold applicable to all images could not be determined. The experimental results presented in Table 3 clearly indicate the ERV thresholds used in corresponding databases.

The DTCWT based background pruning technique as applied to an image from the CMU PIE database and its efficiency in pruning the background can be seen in Fig. 4c. The image with most of the background cropped is processed further.

Observations on profile images show a strong correlation between the locations and distances of various facial features, such as the nose, eyes and ear, from each other. This has been utilized to divide the image into two *approximate* segments, one containing the nose and eyes and the other containing the ear (Fig. 5a). Dividing a profile image of the face into two equal segments, almost always contains the ear in one segment and other features such as the eyes, nose and lips in the other segment. In order to accommodate for pose variances, the image has been split in a ratio of 60% to 40%, the region containing the ear is taken as the larger segment.

4.2. Development of EBPSO algorithm and its application to ear localization

The image containing the ear after segmenting is subjected to the EBPSO algorithm. The algorithm seeks to create an *Entropy Map* of the image and then reduce the search space in order to localize the ear. The successive sections deal with explanation of the basic mathematical framework of entropy, concept and generation of an entropy map, window size estimation algorithm and the EBPSO algorithm for ear localization.

4.2.1. Mathematical framework of entropy

The basics of information and entropy were discussed in Section 3.4 for one dimensional data. We may further extend this to measure the entropy of 2D images (Gonzalez et al., 2012). Our algorithm explores the use of two different entropies, a 2-bit entropy for a logical image and an 8-bit entropy for a grayscale image with 256 gray levels. The 8-bit entropy primarily determines the textural aspect of the ear such as the depth and amount of color present in an image. Thus the value held is dependent on variation in skin color, the amount of illumination and other such considerations. A 2-bit entropy is evaluated on a logical image, such as an edge image, using Eq. (1). The value of a 2-bit entropy determines the randomness and the density of edges in the region under consideration. This 2-bit entropy is unlike the 8-bit entropy in that it is independent of illumination variations, skin color differences and all so far as these variations do not affect the detection of edges.

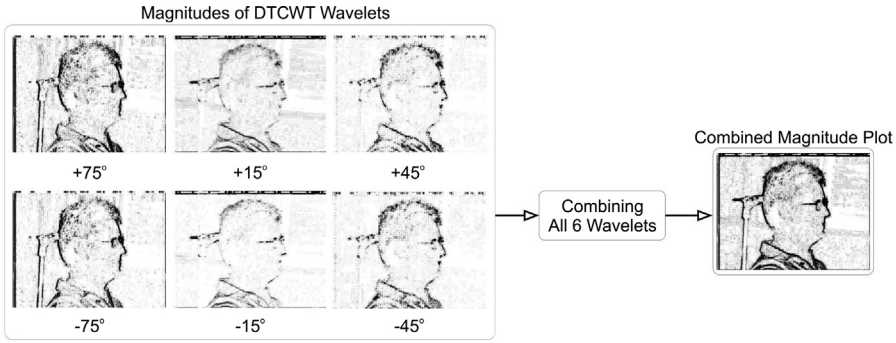

Fig. 3. Magnitude plots (shown as negatives for clarity) of 6 DTCWT wavelets, combining which we get the *Combined Magnitude Plot*.

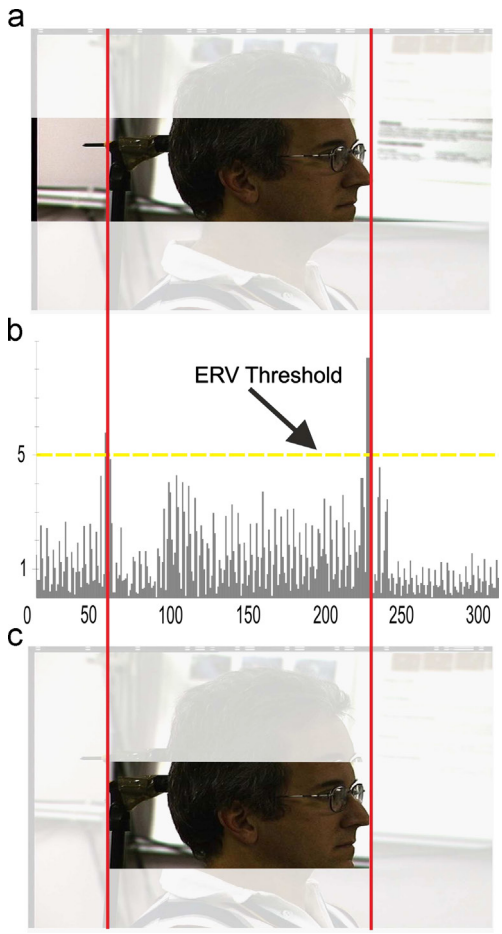

Fig. 4. For a sample Image from CMU PIE: (a) Image returned after skin segmentation, the regions discarded by skin segmentation have been grayed, (b) ERV of the combined magnitude plot, shown in Fig. 3, threshold five is marked in *yellow*. (c) Background has been pruned using the thresholds, the pruned background has been grayed. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

In a binary image, self-information is calculated using a predetermined window size (discussed in Section 4.2.3). As the number of binary ones increases, the probability of ones increases (conversely the probability of zeroes will decrease). Based on their dependency over the total size of the window, larger number of ones translate into a larger value of entropy. Thus we begin to observe that a region possessing maximum entropy has a direct relationship with the distribution of edges and non-edges within a window of specific size.

With the preliminaries of both types of entropies considered, each lends its own value to different sections of our proposed system. The 8-bit entropy is important in window size estimation while the 2-bit entropy is a vital part of generation of the entropy map. The reasons for their use are discussed below.

4.2.2. Concept and generation of entropy map

Entropy map provides a feature set to analyze an image. Here, rich density of edges on the ear is the basic phenomenon on which EBPSO focuses. To use this feature effectively an entropy map is generated. This map is created by placing a window (the size of which is estimated *once* for every database, using a *window size estimation algorithm*, as discussed in Section 4.2.3) on the edge image of Fig. 5b. Fig. 5c depicts the movement of window over this image and the evaluation of entropy values for edges within this window. The movement of the mask and evaluation of entropy together generates the entropy map and can be mathematically modeled by

$$H(image) = -[P(1)_{i,j} \times \log_2(P(1)_{i,j}) + P(0)_{i,j} \times \log_2(P(0)_{i,j})] \tag{4}$$

where

$$i = 1, 2, \ldots, (r-p+1); \quad j = 1, 2, \ldots, (c-q+1);$$

$$P(1)_{i,j} = \frac{\text{No. of 1s within the mask}}{p \times q}$$

and

$$P(0)_{i,j} = \frac{\text{No. of 0s within the mask}}{p \times q}$$

Here, *H(image)* is the entropy map, p is the row width of the mask, q is the column width; r is the number of rows in the complete image and c is the number of columns.

4.2.3. Window size estimation

The size of window plays a principal role in the proposed detection technique and it is used to generate the entropy map. It is also a basis to predict the approximate size of the ear from a given profile image. While it is highly unlikely that the size of the window remains constant over all the databases, it is possible to predict the window size that best fits a given database.

The process flow for this estimation is shown in Fig. 6. We begin by randomly choosing about 10% of the images from the databases. These images are then preprocessed. On the preprocessed image a canny detector is applied and a 2-bit entropy map is created as described in Section 4.2.2. An arbitrarily large window size (slightly less than (1/4)th of the subject's face in the image) is chosen to create the entropy map. From the entropy map the pixel with highest entropy is identified and this pixel's position forms the origin to draw four quadrants. The resulting fourth quadrant (bottom right quadrant) contains the ear, shown by the red markers in Fig. 7a. These coordinates localize the ear within a much smaller region as compared to the original image.

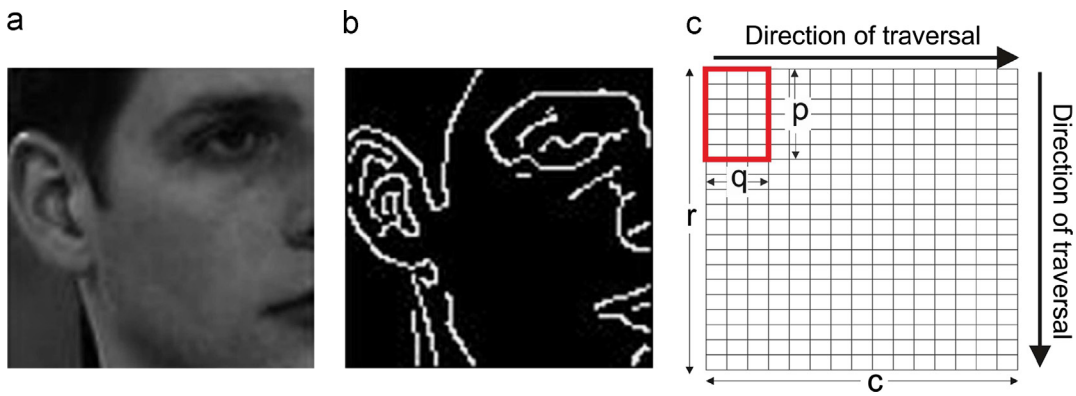

Fig. 5. Steps involved in generation of entropy map. (a) Segmented image. (b) Canny Edge detection. (c) Direction of scan for a mask of size $p \times q$ over the complete image of size $r \times c$.

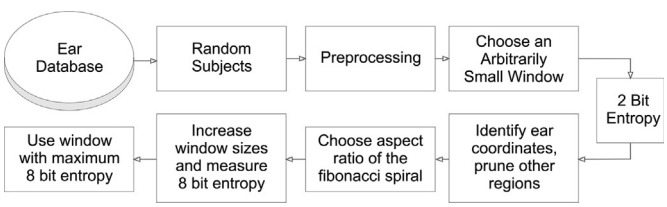

Fig. 6. Process flow of the proposed window size Estimation algorithm.

Fig. 7b shows the segregation of the ear for an image of CMU PIE database. The direction of window traversal allows us to assume that the ear lies in the 4th quadrant of the coordinates. This image is extracted; the ear invariably lies on top left corner of the extracted image as seen in Fig. 7b.

To precisely predict the size of the ear, another window, whose size is deduced by the *ear's discernible resemblance to the Fibonacci spiral* (Fig. 7c), is utilized on the extracted image. The aspect ratio of the window is then maintained constant while changing the row height (or column width) to obtain an optimum row height and column width that encompass most of the ear.

This estimation of window size occurs on a region dominated by the skin texture. Thus the use of entropy of an 8-bit grayscale image was found to be a much better measure as opposed to a 2-bit entropy. This was due to the perceptible depth of skin color within the ear region.

For example, in CMU PIE the subject's face after skin segmentation and background pruning will be 82×82. The width of the ear was found to be greater than 15 pixels. The minimum height of the ear was determined to be 15×1.618, where we assume that the ear height to ear width ratio will be equal to the *Golden Ratio*, i.e., 1.618 (Saraf and Saraf, 2013). This results in a height of approximately 24 pixels. Hence 24×15 is used as the minimum window size.

Observations have shown that the height of the ear appearances in these databases is less variant compared to the width of the ear. Thus the window size is gradually increased by increasing the width and keeping the aspect ratio of 1.618 constant. The 8-bit entropy is calculated for each new window size and the window with maximum entropy value is chosen. This process is repeated for all CMU PIE images chosen and the average window size thereby obtained was 34×23. From here on the window height is kept constant at 34 and the window width is gradually increased in steps of 1 pixel and for each new window size, i.e., 34×23, 34×24, 34×25, etc., the efficiency rate η_R was estimated. The maximum efficiency rate for CMU PIE by this method was found to be 82.50% for a window size of 34×25 with a ERV threshold of 5 (refer Table 3).

4.2.4. Ear localization based on EBPSO

The objective of this section is to reduce the search space provided by entropy map generation. The entropy map is a large dataset and although the actual sizes vary with each database, the search space is still large and localization based on this is difficult. Thus the optimization of search space becomes necessary. In this paper, we propose the use of BPSO (Kennedy and Eberhart, 1997), used as a popular feature selector in a number of previous literature (Ramadan and Abdel Kader, 2009), in the form of EBPSO, a *search space optimizer*. Its outline is given in Fig. 8.

The parameters that define EBPSO are tabulated in Table 1. The BPSO contains 30 particles. Each can be visualized as representing the entire entropy map, as in Fig. 9. Initially all the particles contain randomly initialized ones and zeros. Note that, in a particle, '1' indicates the inclusion of the entropy value at that pixel in the final optimized entropy map and '0' indicates an exclusion. Initially, each of these particles poses a potential solution for the optimization of the entropy map. As they evolve, they choose different sets of entropy values and for each new combination they obtain a new fitness value F:

$$F = \frac{\sum_{i,j=1,1}^{r,c} P(i,j) \times E(i,j)}{\sum_{i,j=1,1}^{r,c} E(i,j)}. \tag{5}$$

Here, $P(i,j)$ is the particle's value at the pixel location and $E(i,j)$ is the corresponding entropy value. The numerator is the sum of all the entropy values chosen by the particle. The denominator is the common normalization factor used to compare the fitness values among the particles.

From the fitness function it can be seen that the particle which chooses pixels with high entropy values and discards pixels with lower entropy values will contain comparatively higher fitness value. The particles iterate to achieve the maximum possible fitness value. After the evaluation of the fitness weight, the particles' velocities are varied. In order to facilitate this, each particle's personal best fitness value and the global best fitness value are stored. The velocity of the particle changes based on these two values. Eq. (6) gives the formula for velocity; it is very similar to the PSO velocity update equation suggested by Rathi et al. (2010). The only change is that a uniform random variable *rand* is multiplied to the previous velocity v_t in order to reduce the rate of convergence of the particles. This prevents rapid convergence and increases the expanse of global search performed by the particles:

$$v_{t+1} = v_t \cdot \omega \cdot rand + c_1 \cdot (1 - t/t_{max}) \cdot (P_{N_{best}} - x_t) + c_2 \cdot (t/t_{max}) \cdot (G_{best} - x_t) \tag{6}$$

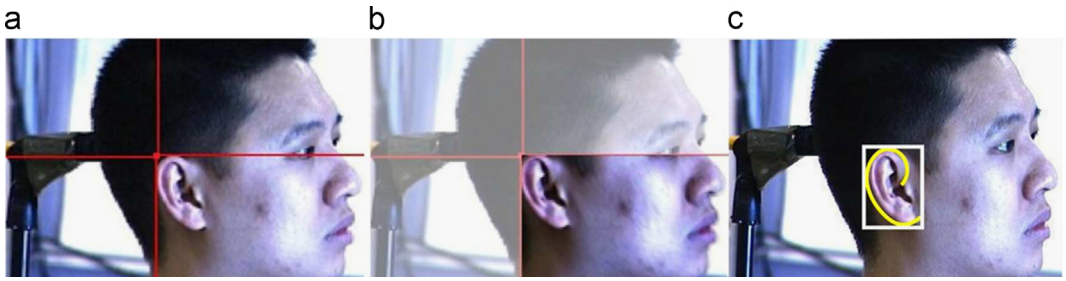

Fig. 7. Process of localization: (a) Partitioning into four quadrants, (b) Ear present in the 4th quadrant, (c) Close resemblance of ear and the fibbonacci spiral (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

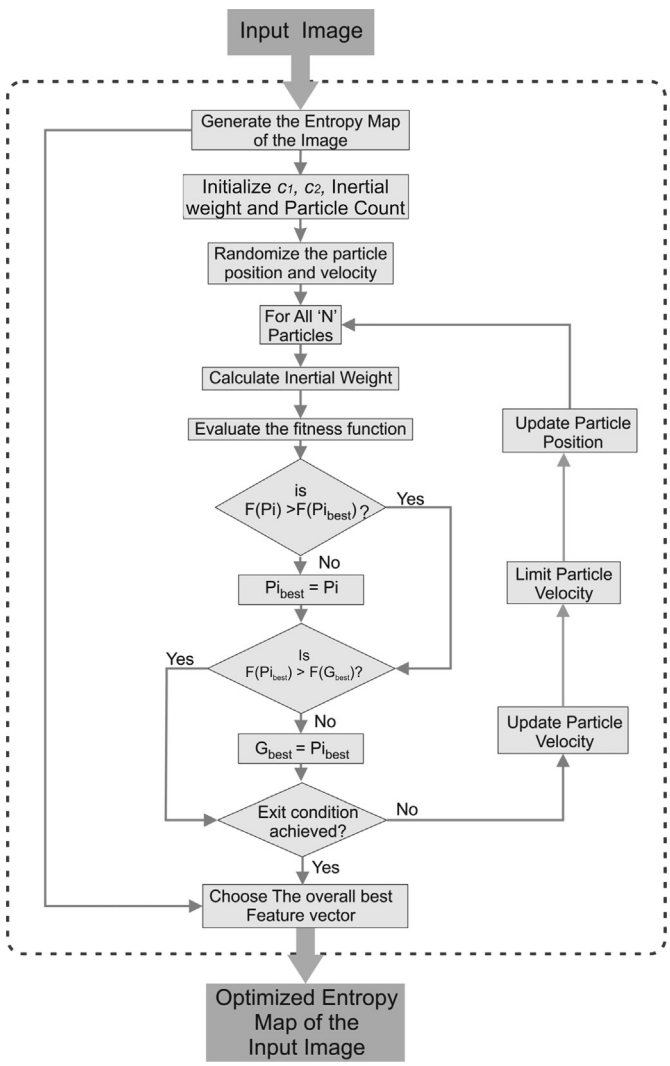

Fig. 8. Flow chart of the proposed EBPSO algorithm; the notations used are similar to those used by Ramadan and Abdel Kader (2009).

Table 1
Parameters utilized in EBPSO.

Parameters	Values
Maximum iterations (t_{max})	20
Inertial weight (ω) (maximum)	0.9
Inertial weight (ω) (minimum)	0.4
Self-learning factor (c_1)	2.0
Social-learning factor (c_2)	2.0
Particle count (N_{max})	30
Maximum velocity	2.5

In Eq. (6), velocity is updated based on iterations. $P_{N_{best}}$ and G_{best} are the personal best and global best values, while x_t is the previous position value. BPSO suggests mapping the velocity between [0, 1].

For this, a sigmoid function is used as below:

$$S(v_{t+1}) = \frac{1}{1+\exp(-v_{t+1})} \tag{7}$$

Once the velocity is obtained, the position of *each pixel* in the particle is updated as in the following:

$$x_{t+1} = \begin{cases} \overline{x_t} & \text{if } rand \le S(v_{t+1}) \\ x_t & \text{otherwise} \end{cases} \tag{8}$$

where $\overline{x_t}$ represents the 1's complement of x_t.

After each particle is evaluated by the fitness function given by Eq. (5), the particle with highest weight at the end of the optimization is assumed to contain the best entropy locations for the search. These iterations are continued until either of the exit condition mentioned below is encountered:

1. If the fitness value remains the same after 5 consecutive iterations, or
2. If a maximum of 20 iterations are completed.

Optimizations using EBPSO generate one particle, which is used to eliminate unnecessary pixels from the original entropy map which in turn helps generate a new optimized entropy map with significantly reduced search space. 3D and planar visualizations of the same are shown in Fig. 10a and b. The two versions of the entropy map allow us to infer the following:

1. In the 2D planar entropy map the region corresponding to high entropy is highlighted in red, while the regions with lower entropy values are blue in color. This shows that the density of edges decreases as the mask moves away from the ear and the entropy also gradually reduces as distance from the ear increases. This clearly shows that the regions around the ear invariably contain high entropies providing a basis for accurate detection and classification.
2. The 3D surface plot shows the subtle variations in peaks around the region that may contain the ear. Here the pixel corresponding to the highest entropy value will be the ideal location to place the window to localize the ear.

On the entropy map, *a pixel-wise search for the highest entropy value* is performed. Owing to the optimizations by the EBPSO, the search conveniently discards pixels eliminated by EBPSO and localizes to a pixel from the remaining region. Thus a search on 2784 pixels on the original entropy map is reduced to a search only on 1464 pixels by the EBPSO, as shown in Fig. 10b. In ear localization we identify the coordinates containing the maximum entropy values. The point corresponding to these coordinates is *extrapolated onto the*

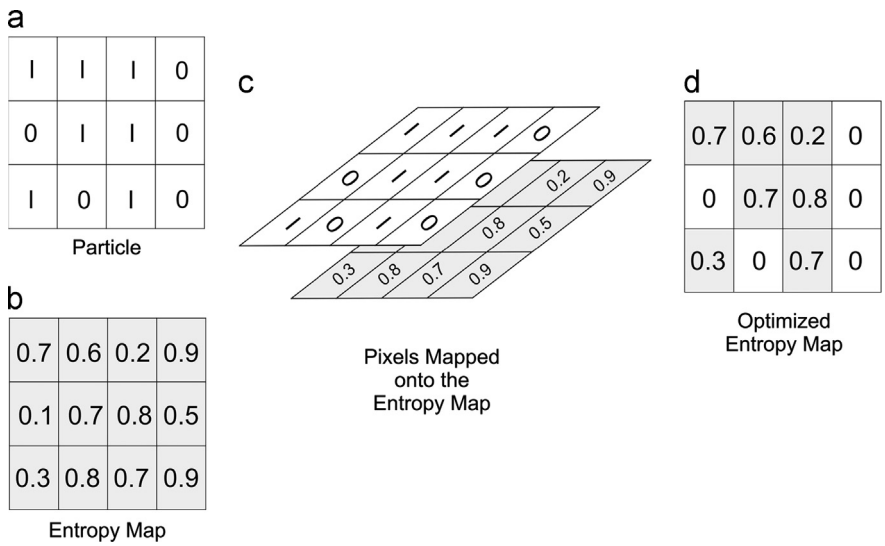

Fig. 9. A visual representation of particles and the corresponding entropy map; note how the pixels with zeros in particle were utilized to remove the corresponding entropy values in the optimized entropy map. The combination of 1's and 0's in a particle is also referred to as a feature vector.

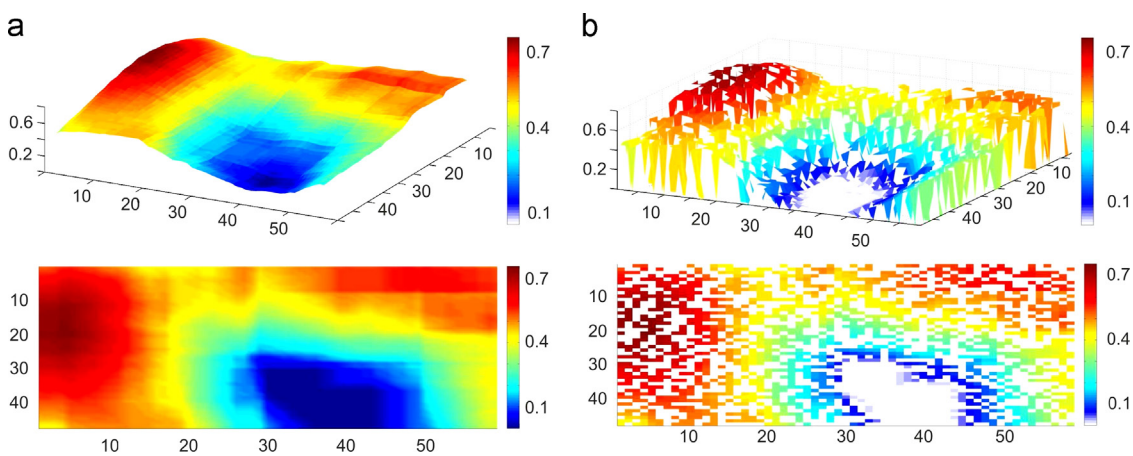

Fig. 10. Visualizations of search space before and after EBPSO Optimizations. (a) Original entropy map of Fig. 5b in 3D and planar representation (*Features before EBPSO* = 2784). (b) Optimized entropy map in 3D and Planar representation (*Features after EBPSO* = 1464). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

original image and used as the reference point (top left corner) to obtain the window with maximum value of entropy. Through this technique the window is found to contain the ear.

After identifying the region considered to possess the ear, the correctness of detection is verified through classification.

4.3. Entropy based classification

Techniques like SVM or other such methods may be used for classification. However, considering the variations in size and conditions in which databases were taken, these will require a lot of training to provide accurate measurement.

Over the course of experimentation, a general trend in the values of 2-bit entropy in all databases was observed. Regions not containing the ear had lower entropy values in comparison to the regions which contained the ear. The window containing the ear consistently provided values of entropy above 0.5 and below 0.8, albeit with a few obvious exceptions. Thus a threshold, closer to the lowest value of entropy held by an ear segment, can be used. For example, Fig. 11 shows the segregation of ear and non-ear regions with a threshold of 0.62 in CMU PIE database. For entropies below this value it is assumed that the window does not contain the ear. These values, although varying subtly with the databases, present a predictable measure for classification. It is also much simpler to implement compared to other classifiers.

5. Experimental results and discussions

This section evaluates the performance of the proposed ear detection system. For the sake of experimentation, four databases have been chosen; Table 2 lists these databases and highlights the challenges posed by each. The profile of the images from the databases is assumed to be known to system. Experiments were conducted on MATLAB (2012).

The performance metrics used to measure the efficiency of the proposed system are Detection Rate, False Acceptance Rate (FAR), False Rejection Rate (FRR). These are standard metrics that have been used in several literature previously. A brief description of each is given below:

1. *Detection Rate*: It is the ratio of the number of images in which ear has been identified to the total number of images. A threshold based on entropy values, as discussed in Section 4.3, is used to perform this classification.
2. *False Acceptance Rate (FAR)*: It is of the ratio of the number of non-ear regions falsely detected as an ear region over all the

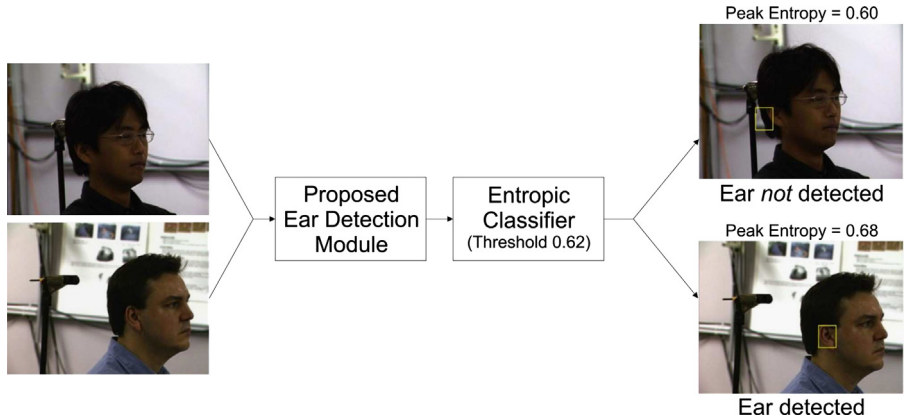

Fig. 11. Use of 0.62 as threshold for entropy values in CMU PIE database to segregate ear and non-ear regions.

Table 2
Details of images from four databases used for experimentation.

Database	Image type	Image characteristics	No. of subjects	Images per subject	Total Images
CMU PIE	RGB	Cluttered background and varying illumination	30	8	240
Pointing Head Pose	RGB	Pan and Tilt variations	12	44	528
FERET	RGB	Images procured at different distances from camera	43	6	258
UMIST	Grayscale	Minor pose variations and minor occlusions	15	16	240

images present. In this case non-ear segments have higher entropy than the ear segment and the value of entropy is greater than the classifier threshold, hence they are wrongly considered as ear regions.

3. *False Rejection Rate (FRR)*: It is the ratio of the number of ear regions falsely rejected as a non-ear region over all the images present. In this case ear segments possess entropy lesser than the classifier threshold and are erroneously considered as non-ear regions by the classifier.

Although the above performance metrics served to highlight certain basic features of the results obtained, individually they were inadequate in determining the ideal combination of Window Size and ERV threshold value for a given database. With reference to Table 3, for Pointing Head Pose Database, the second entry is a good example. Here, the highest detection rate (88.64%) occurs for a threshold of four and window size of 29×17, the FAR value here is 7.20%. However the actual least value of FAR in the overall table occurs at a threshold of 6; at this threshold, however, the detection rate is too low (only 42.05%). Thus although Detection Rate, FAR and FRR served to highlight the functioning of the system, using them individually, it is very difficult to obtain the ideal parameters (ERV threshold and window size) for a given system. This necessitated the development of a new metric.

Apart from using the above metrics, we also manually computed the effective number of ears detected, and this was independent of the classifier identifying the detections as successful or unsuccessful. Doing so gave us a parameter, which we have termed as the *efficiency rate* (denoted by η_R). With this we could analyze the system, determine its performance and the ideal combination of ERV threshold and window size. Incidentally, we found that this value can be obtained using the Detection Rate, FAR and FRR, and is calculated using

$$\eta_R = \text{Detection Rate} + \text{FRR} - \text{FAR} \tag{9}$$

Adding Detection Rate and FRR gives the total number of ears detected along with the false detections, then by subtracting the FAR all the falsely detected ears are accounted for. Thus the Efficiency Rate (η_R) measures the actual number of ears detected. η_R is independent of the working of the classifier since it depends only on the actual number of ears detected. Further for any given database it highlights a balanced combination of ERV threshold and window size so as to get the maximum detection rate compensating for issues with the classification used.

The choice of the entropy value for classification was based on experimental observation. It is not reasonable to use one entropy value for classification in all databases due to variety of images present. However, within a given set of images, it is possible to obtain a unique entropy value to classify the images. Note that, if the entropy value used for classification is extremely low, the detection rate will increase and so will FAR; FRR will fall by an equivalent amount (and vice versa for an extremely large entropy value). When these modified values are used in Eq. (9), η_R value will remain unchanged.

For each database the results have been presented in a tabular format with the best results highlighted in bold. Due to considerable variances in each database, determination of a unique ERV threshold is impractical. Hence, three thresholds have been used for the databases under experimentation. For each threshold, Detection Rate, FAR, FRR and η_R are computed. The results have been presented for five window sizes. The height of the window is constant and the width varies in steps of one pixel (Section 4.2.3 had discussed this in detail).

This experimental Section is divided into seven subsections. The first four subsections present details of the database, the experiments performed, the results obtained and their interpretations. Section 5.5 gives a performance comparison with a standard benchmark as well as the computational complexity of the algorithm. Section 5.6 discusses the general trends observed in the results, successes and failures over all databases. Section 5.7 gives a perspective of our system with respect to ear biometrics.

5.1. CMU PIE

The CMU PIE (2000) database is characterized by the presence of rich background details and several illumination conditions. This makes it an ideal choice for testing the efficiency of the proposed system when applied to realistic surveillance-like environments.

Table 3
Experimental results.

	Window size	Detection Rate in %	FAR in %	FRR in %	η_R in %	Detection Rate in %	FAR in %	FRR in %	η_R in %	Detection Rate in %	FAR in %	FRR in %	η_R in %
ERV threshold →			4				5				6		
	34 × 23	95.83	22.92	2.50	75.42	95.00	19.58	2.92	78.33	90.83	21.67	3.75	72.92
	34 × 24	96.25	22.92	2.92	76.25	95.00	18.33	3.33	80.00	91.67	19.17	2.50	75.00
CMU PIE	**34 × 25**	93.75	22.92	4.58	75.42	**92.92**	**15.42**	**5.00**	**82.50**	89.17	19.17	5.00	75.00
	34 × 26	92.50	21.25	5.83	77.08	92.08	16.67	5.42	80.83	87.50	19.17	5.83	74.17
	34 × 27	91.25	22.50	7.08	75.83	89.58	18.75	7.92	78.75	85.83	19.58	5.83	72.08
	29 × 16	88.64	8.14	1.89	82.39	73.11	8.71	2.08	66.48	42.23	5.49	2.46	39.20
	29 × 17	**88.64**	**7.20**	**2.46**	**83.90**	72.35	7.95	2.46	66.86	42.05	5.11	2.65	39.58
Pointing head pose	29 × 18	88.07	8.90	3.03	82.20	72.16	8.90	2.65	65.91	41.48	5.11	2.65	39.02
	29 × 19	87.50	8.52	3.60	82.58	71.78	7.77	2.84	66.86	40.91	4.17	3.60	40.34
	29 × 20	85.98	8.33	4.92	82.58	69.51	7.77	4.73	66.48	39.96	4.73	4.17	39.39
ERV threshold →			3				4				5		
	30 × 20	94.57	7.75	1.55	88.37	96.12	9.69	0.39	86.82	94.96	13.95	0.78	81.78
	30 × 21	94.57	5.81	1.55	90.31	96.12	10.08	0.78	86.82	94.96	14.34	0.78	81.40
Color FERET	**30 × 22**	**94.57**	**5.43**	**1.55**	**90.70**	95.35	10.08	1.16	86.43	94.19	13.57	1.55	82.17
	30 × 23	94.19	6.98	1.94	89.15	94.96	10.47	1.16	85.66	94.19	14.73	1.16	80.62
	30 × 24	93.80	7.75	2.33	88.37	94.96	9.69	1.16	86.43	94.19	13.95	1.55	81.78
	48 × 30	90.83	13.75	0.83	77.91	**91.67**	**15.00**	**1.25**	**77.92**	91.67	15.80	1.25	77.10
	48 × 31	90.83	14.17	0.83	77.49	91.67	15.42	1.25	77.50	91.67	15.00	1.25	77.90
UMIST	48 × 32	90.00	14.58	1.67	77.09	90.83	15.42	1.67	77.08	89.58	16.70	2.92	75.80
	48 × 33	88.75	14.58	2.92	77.09	88.75	15.42	2.92	76.25	89.17	17.50	3.33	75.00
	48 × 34	86.67	14.17	5.00	77.50	87.50	15.83	4.58	76.25	87.50	16.30	4.58	75.80

Table 2 indicates that a total of 30 subjects has been used and 8 profile images are chosen per subject. This results in a custom database of 240 images. 8 profile images for every subject constitutes two images with ambient lights turned off, two images with enhanced lighting, two with ambient lights turned on and finally two more with a different set of background. Sample images from the database are shown in Fig. 12a.

The ERV threshold for background pruning was empirically determined to vary between 4 and 6 for images from this database; hence the detection rates are computed for three threshold values, namely 4, 5 and 6. The initial window size was determined, in accordance with Section 4.2.3, to be 34 × 23. The window size is varied between 34 × 23 and 34 × 27 in steps of one pixel. Also, an entropy value of 0.62 was used to classify ear and non-ear segments. The results are tabulated in Table 3, the η_R values are plotted against window sizes for all three ERV thresholds and can be seen in Fig. 13a.

Based on the tabulations (Table 3), and the plot of η_R vs. window size (Fig. 13a) over all three thresholds, we observe that a threshold of 5 provides consistently high values. This signifies that 5 is the ideal ERV threshold for background pruning. The highest η_R of 82.50% occurs at window size of 34 × 25.

As noted previously, the CMU PIE database is characterized by cluttered background and variable illumination conditions. The promising results can be attributed to two key concepts developed above. Firstly, the effectiveness of the background pruning technique plays a critical role in eliminating most of the background. Secondly, the usage of 2-bit entropy instead of a 8-bit entropy plays a crucial role in handling poor illumination conditions. Since 2-bit entropy considers only the edge density, the skin color variations that occur due to poor illumination conditions are not accounted for. Thus the EBPSO based on 2-bit entropy map works very well even under such arduous conditions.

5.2. Pointing head pose database

The Pointing Head Pose (2004) database has images of 15 people with 93 images per person at different poses. The pose (or head orientation) is determined by two angles, namely horizontal and vertical.

In horizontal direction, angles include −90°, −75°, −60°, −45°, −30°, −15°, 0°, +15°, +30°, +45°, +60°, +75° and +90°. Negative angles are left profile images and positive angles are right profile images, 0° is frontal image. In vertical direction, the angles are −90°, −60°, −30°, −15°, 0°, +15°, +30°, +60° and +90°. Here the negative sign indicates downward facing and the positive sign indicates upward facing images.

In order to use this database for ear detection, a custom database was created which included images with all vertical angles between −60° and +60° and horizontal angles ±90°, ±60°, ±45° and ±30°. We discarded images where ear was completely occluded by hair. This resulted in a database with images of 12 people with 44 poses each. Sample images from our database with their respective pose angles are shown in Fig. 12b.

The thresholds used for background pruning are same as those of CMU PIE (4, 5 and 6). The entropy value used for classification is 0.6. The window size was determined to be 29 × 16 with constant height, and width incrementing in steps of one pixel.

With regard to the results tabulated in Table 3 and the plot of η_R versus window size shown in Fig. 13b over all three thresholds, we may conclude that an ERV threshold of 4 for background pruning provides consistently high values. The highest Efficiency Rate of 83.90% occurs at window size of 29 × 17.

It can be seen on the plot of Fig. 13b that the η_R values for a given ERV threshold are very consistent. The proposed ear detection system uses the measure of entropy to detect the presence of ear in a window of specific size. Due to the *mathematical nature of entropy*, it ignores the geometric aspects of the ear's shape. Thus, regardless of the orientation of the subject's ear, the proposed system accurately detects the ear features, discussed again in Section 5.6. However, if the background pruning fails to correctly identify the subject's facial features, the localization of ear which follows also fails. This causes a drop in η_R values at ERV thresholds of 5 and 6.

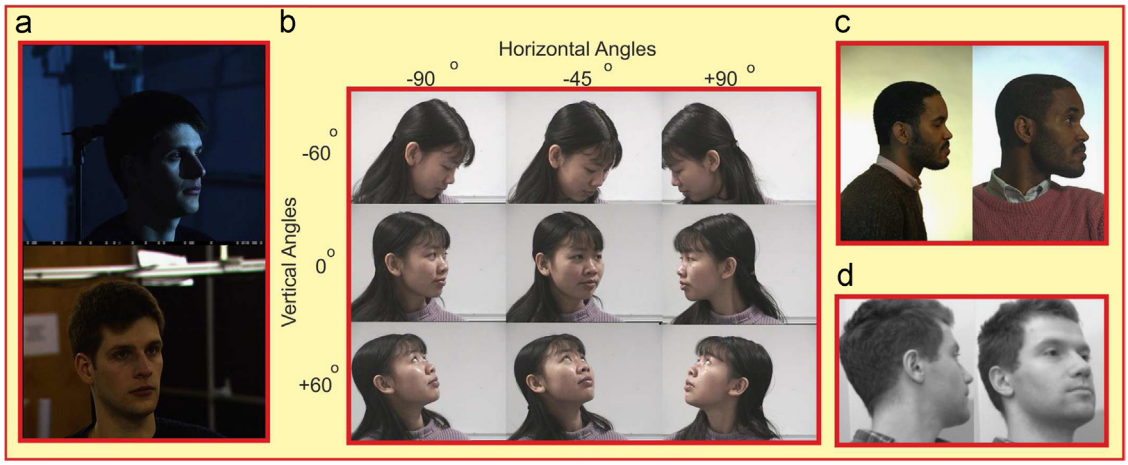

Fig. 12. Sample images from all databases used.

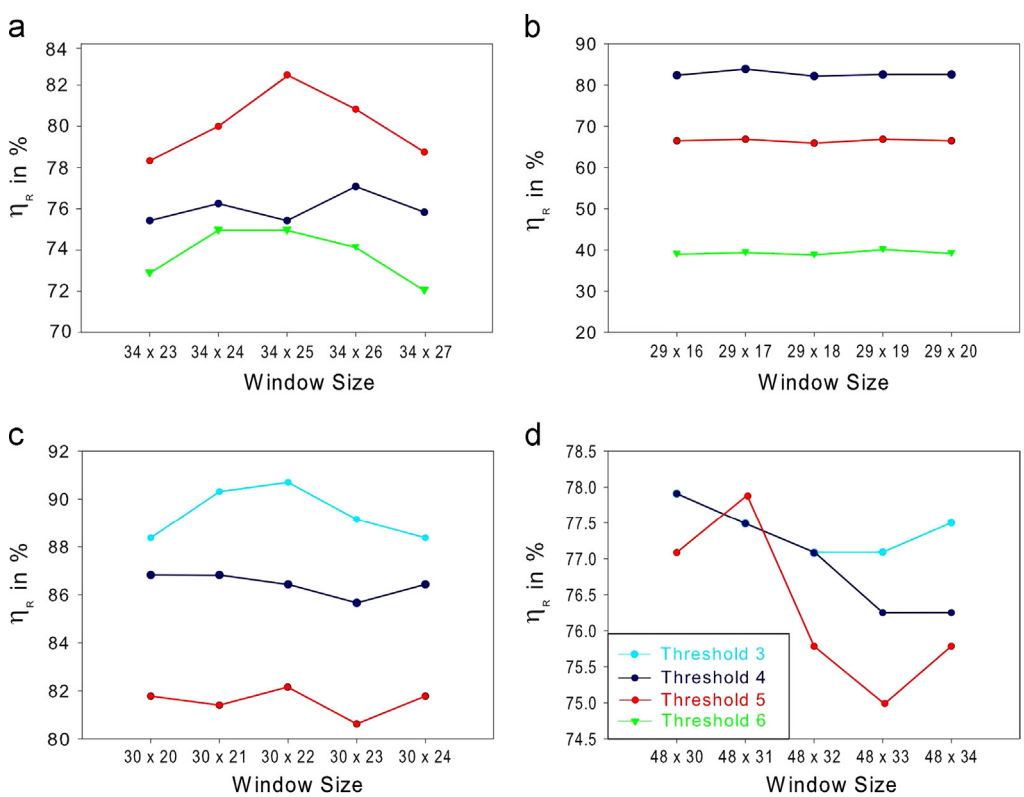

Fig. 13. Graphical representation of Efficiency Rate vs. window size.

5.3. Color FERET

The Color FERET (2003) database contains 14,126 images in 20 pose angles varying from profile left (-90°) to profile right ($+90^\circ$). For this experiment, only pose angles of $\pm 90^\circ$, i.e., profile right and left, were chosen. Among all subjects, only the ones which had their images taken at various distances from the camera were chosen. Also, subjects whose ear was completely occluded were not chosen. This yielded a database with 43 subjects under six different poses, totalling to 258 images. Sample images are shown in Fig. 12c.

For this customized database, the ERV threshold for background pruning was determined to lie between 3 and 5 (thus values 3, 4 and 5 were used). For classification, an entropy value of 0.56 seemed ideal as it incurred the least FAR and FRR values.

With reference to results tabulated in Table 3, and the plot of η_R versus window size shown in Fig. 13c, over all three thresholds, an ERV threshold of 3 for background pruning is seen to provide consistently high values. The highest Efficiency Rate of 90.70% occurs at window size of 30×22.

The results signify the effectiveness of the proposed ear detection system in handling images with varying ear sizes. As previously discussed toward the end of Section 4.1.2, the profile image is split into two approximate segments in a 60% to 40% ratio (the larger segment containing the ear); while doing so we assumed that, by knowing the locations of facial features in a profile image, we may approximately predict where the ear lies. This prediction and subsequent segmentation work very well whatever size the profile image may be, and the results obtained for FERET database substantiate this.

5.4. UMIST

The UMIST (1990) database, currently known as the Sheffield Face Database, consists of 564 grayscale images of 20 individuals of mixed race, gender and appearance. Each individual is shown in a range of poses from profile to frontal views. For ear detection purposes, a subset of the original database consisting of 15 subjects with 16 images each, totalling to 240 images was created. While defining this customized database, fully frontal images as well as ones with severe hair occlusions have not been taken into consideration. Sample images of this database are shown in Fig. 12d.

Like in FERET, the threshold values used for background pruning are 3, 4 and 5. The entropy value used for classification is 0.64. The window size was determined to be 48×30.

Table 3 and the graph in Fig. 13d show the results obtained for this database. The highest Efficiency Rate of 77.92% occurs at a window size of 48×30 and an ERV threshold of 4.

The main purpose of using this database is the presence of grayscale images. Although skin segmentation plays a key role in the proposed system, for UMIST, the skin segmentation was not performed and the proposed system still works giving promising results. This shows that the overall system is perfectly capable of handling images without the integral skin segmentation block.

5.5. Comparative study

The three techniques mentioned in Section 1 (Islam et al., 2008; Wu et al., 2008; Abaza et al., 2010) are characterized by large times in the training stage of the algorithm. Successful alterations to the initial algorithms have yielded an approximate time of 8 hours as stated by Abaza et al. (2010). Our algorithm requires 8.71 seconds per image (from CMU PIE), in order to find the approximate window size for each database. Further the training stage in other algorithms focuses on teaching the difference between ear and non-ear regions, while our algorithm is trained to estimate the dimension of the ear in a given database.

The strength of the technique proposed by Prakash and Gupta (2012) is derived from the successful detection of skin regions. Similarity of the hair color with skin color in UND database reduces the performance of skin segmentation and, in turn, affects the ear localization accuracy. In this paper, skin segmentation is used only as a preprocessing technique, and the ear detection does not depend on it directly. Instead, our method works successfully irrespective of the hair color. Additionally, one of the key causes of failure in Prakash and Gupta (2012) can be attributed to noise and poor illumination. But we use an entropy based approach, and the ear detection is successful even in very poor illumination conditions, as is evidenced by the results from CMU PIE database. Lastly, in Prakash and Gupta (2012) the template used for matching is required to exhibit scale, rotation and illumination invariant characteristics for proper ear detection; the system's accuracy depends on the number of training images. The proposed method places no such restrictions on training images. In fact, the training images are chosen randomly.

A quantitative performance comparison of the proposed technique could not be made with the references mentioned above because these techniques have used UND and IITK databases which were not used by us. However, these databases are easier to work with in comparison to the databases we use. These databases (UND and IITK) contain images captured in consistent background under controlled illumination conditions which are tailored for ear detection purposes. Nonetheless, a detailed benchmark study in terms of a qualitative performance comparison with existing technique, as provided in Table 4, shows the promising performance of the proposed technique.

After the window size has been estimated, the detection itself requires an average time of 12.18 s. The time was obtained on personal computer with Intel® CORE™ i5 (2.53 GHz) and 4 GB RAM.

5.6. Discussions

Over the course of all experiments conducted, the following trends are observed:

1. The variations in η_R, FAR and FRR values, over different ERV threshold values within the same window size indicate the direct impact of DTCWT based Background Pruning over the functioning of the complete system. If the threshold used is either higher or lower than the ideal threshold, background pruning fails and so does the ear localization. This is evident in almost all databases used.
2. Analysis of graphical data and the tabulation reveals that, for a given ERV threshold, η_R values are not monotonic and they are subject to slight variations. However, as previously noted, η_R values drop significantly (by 10–30% depending on the database) at non-ideal ERV thresholds.

The databases are chosen to negotiate specific challenge(s) as shown in Table 2. In Fig. 14a, two important cases of success in CMUPIE have been shown. The first image shows the ability of our algorithm in overcoming poor lighting conditions. Given a situation of poor lighting condition in CMU PIE, there is a reduction in

Table 4
Comparison of the proposed technique with the technique in Prakash and Gupta (2012).

Parameters	Techniques	
	Prakash and Gupta (2012)	Proposed technique
Detection time	7.95 s	12.18 s
Training	A few 100 samples.	**Very less. About 20–50 samples.**
Invariance to		
(i) Illumination	Upto some extent	**Yes**
(ii) Rotation	Yes	Yes
(iii) Scale	Yes	Yes
(iv) Color and grayscale Image	No	**Yes**
(v) Occlusion	No	**Minor occlusion**
(vi) Both profiles	Yes	No
(vii) Template matching	Yes, using SURF	**No**
(vii) Background	Little background	**Highly cluttered**
Test dataset	Considerable scaling and rotation	**Significant scaling, pose, illumination and background clutter**

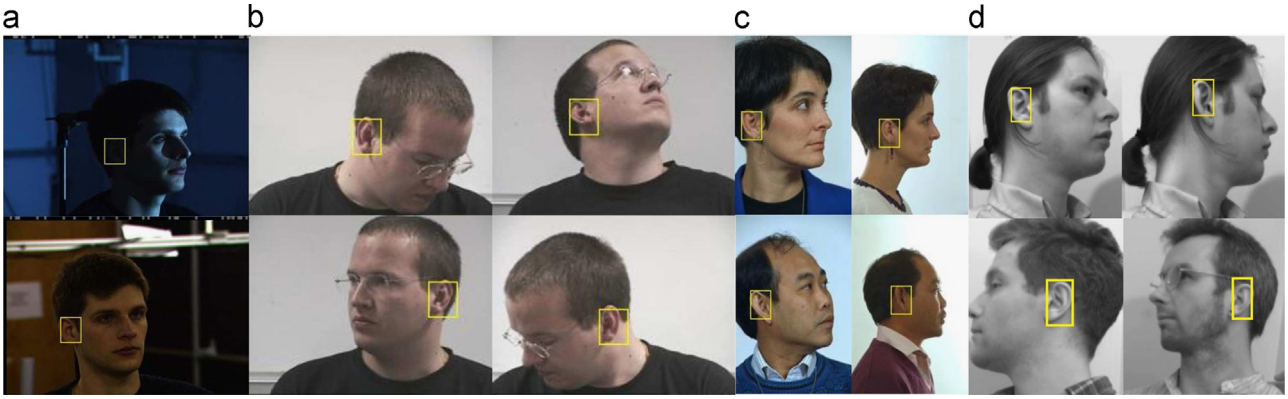

Fig. 14. Illustrations of successful detections. Images from (a) CMU PIE with illumination and background variations, (b) Pointing Head Pose with varying pan and tilt, (c) FERET with scale variations and (d) UMIST database which are grayscale.

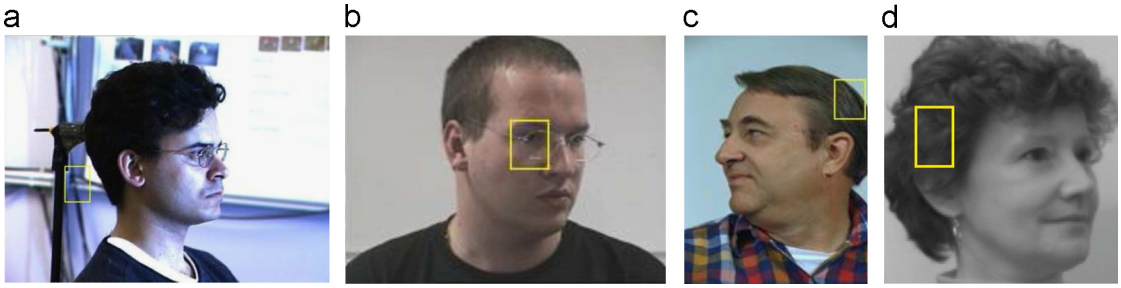

Fig. 15. Cases in which the detection technique fails.

intensity of edges in the overall image. Approximate segmentation of the image after background pruning removes all the other regions rich in edges like the nose tip, eyes, etc., and compared to the regions surrounding the ear itself, ear will still possess the maximum number of edges. Thus entropy is successful in capturing the ear even in poor illumination. The second image in Fig. 14a shows detection in the presence of rich background details. The application of DTCWT based background pruning is a primary contributor in successfully identifying the subject's facial features and removing the background.

Since entropy is a mathematical measure with no relation to the geometric aspects of the ear, this plays a crucial role in Pointing Head Pose Database. This database was specifically chosen due to immense pan and tilt variations. Here although the ears' position and orientation varied immensely, their structures were still visible to the naked eye. With entropy being independent of the orientation of the ear, its application over Pointing Head Pose produced successful detections, as shown in Fig. 14b.

Fig. 14c shows the performance under the impact of scale variance, in FERET database. During preprocessing, after DTCWT based background pruning, the image is divided into two regions in a 60% to 40% ratio with the larger segment containing the ear. In FERET database, this division plays a critical role in successful detection.

Validation of our algorithm over grayscale images using UMIST database yielded convincing results. This is due to the presence of necessary properties of the ear required for detection (Fig. 14d).

Beyond these cases of successful detection, there are specific failures which have been discussed next.

There exist certain images in CMU PIE where the heavily detailed objects in the background interfere with background Pruning. In such cases, ERV thresholds yield incorrectly extracted images, containing both the subject as well as a fair bit of background. This has a detrimental effect on the stages which follow, causing failure in ear detection. An example of this is shown in Fig. 15a.

A prominent cause of failure in Pointing Head Pose Database is that of reduction in visibility of the ear. When the physical structure of the ear is not clearly visible in the image, due to severe pan and tilt variations, applying edge causes other regions to show well defined edges as compared to the ear. Thus, EBPSO localizes upon these regions. In Fig. 15b the person's spectacles are detected.

A common cause of failure in both FERET and UMIST is the extensive presence of features around the ear. Due to this, the density of edges observable is more in other regions of the subject, specifically the hair or in specific cases that of the nose itself (as a result of incorrect segmentation). In Fig. 15(c) and (d) for subjects from FERET and UMIST, respectively, the hair region gets detected as ear region.

5.7. Toward ear biometrics

Stages involved in the development of a basic biometric system based on human ear shape is shown in Fig. 16. In accordance with this, we see that ear detection is the foremost and a very essential stage. It aims to identify and extract the ear region in a given profile image of a person. Ear detection is an essential step in this biometric system because failures here will surely undermine the system's utility. Even though current ear detection and recognition systems have reached a certain level of maturity, their application and success is limited to several constraints like controlled environment, variations in pose, etc.

While research focuses on ear recognition, the benefits of using ear as a biometric cannot be realized unless the systems can work in environments that are realistic and be able to handle variety of effects an image may be subject to. In an effort to improve current systems, we have undertaken the development of ear detection. With the capability of our detection technique in handling a large variety images under various conditions, we anticipate a significant

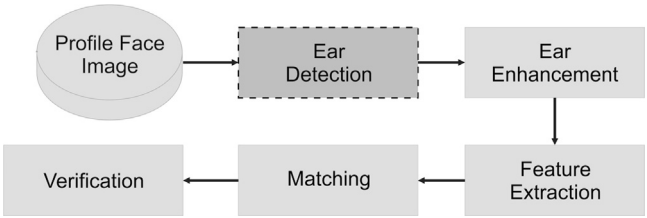

Fig. 16. An overview of ear biometric system.

improvement in the recognition stage and thus the biometric system as a whole.

6. Conclusions

A novel approach for a flexible ear detection system in facial images is proposed, which uses EBPSO to reduce search space in analyzing entropy and in localizing the ear. DTCWT based background pruning has played a key role in efficient background removal. A successful attempt has been made to equally handle all face image variations (pose/background/illumination) which undermine ear detection. Entropy is successful not only in localizing ear but also in classifying ear and non-ear regions.

The experimental results on various benchmark face databases indicate that the proposed method has performed well under severe illumination and background variations with η_R having reached 82.50% for a window size of 34×25 and an ERV threshold of 5 for CMU PIE face database. Thus the proposed method s efficient in practical situations where the facial images may be taken in uncontrolled and unknown surroundings. It is also successful in tackling the most challenging task of detecting ear from facial images with severe pose and occlusion variances, with η_R of 83.90% for Pointing Head Pose database, 90.70% for Color FERET database, and 77.92% for UMIST database. Thus, the proposed method has proven to be a promising technique under arbitrary variations in illumination, poses and backgrounds with occlusions too.

The possibility of using EBPSO as a feature extractor and implementation of a profile identification module prior to ear detection is being studied. This work is currently under progress.

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